

A fractional-dimensional Moser–Trudinger inequality for real curvature dimension

Solution packet for the question in arXiv:2506.15621

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Positive answer in the source’s geometric regime. For every real $N > 1$, there is a natural Moser–Trudinger inequality on infinite-volume $\text{CD}(K, N)$ spaces with positive unit-ball volume and positive Cheeger constant. The critical exponent is $N/(N - 1)$, and the exponential is truncated after precisely the powers that need not be controlled by L^N . The proof is an isoperimetric rearrangement followed by Moser’s one-dimensional lemma; neither step requires N to be an integer.

The question and the natural fractional formulation

Bronstein defines $(MT)_\alpha^n$ for integer $n \geq 2$ and asks whether $\text{CD}(K, N)$ spaces with non-integer N admit a finer version [1].

4. APPLICATION TO THE MOSER–TRUDINGER INEQUALITY

In this section, we consider the following Moser–Trudinger type inequality:

Definition 4.1. Let (X, d, μ) be a metric measure space of infinite volume, $\alpha > 0$ and $n \geq 2$ an integer. We say that (X, d, μ) satisfies the Moser–Trudinger inequality $(MT)_\alpha^n$ if there is a constant $C > 0$ such that, for any $u \in W^{1,n}(X)$:

$$(4.1) \quad \int \text{lip}(u)^n \leq 1 \Rightarrow \int \exp\left(\alpha u^{\frac{n}{n-1}}\right) - P_n(\alpha u^{\frac{n}{n-1}}) d\mu \leq C.$$

where $P_n(t) = \sum_{j=0}^{n-2} \frac{t^j}{j!}$.

Remark. Since X has infinite volume, in general there is no embedding $W^{1,n}(X) \rightarrow L^p(X)$ for $p < n$, hence the necessity of the polynomial correction term. In this whole section, the notation u^r is used to denote $|u|^r$, unless otherwise specified.

Remark. There exist $\text{CD}(k, n)$ -spaces for n not an integer. It is natural to ask whether they would admit a finer version of the Moser–Trudinger inequality for non-integer n , but this is not in the scope of the paper here.

Satisfying such an inequality implies a nice geometry at small and large scale of the space. That geometry will be encapsulated by two quantities, the Cheeger constant and the asymptotic growth ratio, see definitions 2.17, 2.21. Denote also by λ_p the spectral gap of the p -laplacian. Recall that by results of Buser, Matei, λ_p is controlled by the bound on the curvature and the Cheeger constant for manifolds. We will prove here the following result:

Theorem 4.2. Let (X, d, μ) be a locally compact geodesic $\text{CD}(k, n)$ -space with $\mathcal{V}(X) > 0$, and nonzero Cheeger constant. Let n_X denote its isoperimetric constant and $l^{(m)}$ the m -th isoperimetric ratios.

Then (X, d, μ) satisfies the Moser–Trudinger inequality $(MT)_\alpha^m$ for $m \geq n_X$ and $\alpha \leq m(l_\infty \omega_{m-1})^{\frac{1}{m-1}}$. Moreover, (X, d, μ) does not satisfy the Moser–Trudinger inequality $(MT)_\alpha^n$ for $\alpha > n(\theta_\infty \omega_{n-1})^{\frac{1}{n-1}}$.

Remark. In the case of noncollapsed $\text{RCD}(k, n)$ -spaces, $n_X = n$ and $\theta_\infty = l_\infty$, so we have a complete answer to which Moser–Trudinger inequalities are satisfied.

Fix a real $p > 1$ and put

$$q = \frac{p}{p-1}, \quad k_p = \lceil p-1 \rceil, \quad \mathcal{E}_p(t) = e^t - \sum_{j=0}^{k_p-1} \frac{t^j}{j!}.$$

When $p = n$ is an integer, $k_p = n-1$ and $\mathcal{E}_p = e^t - P_n(t)$ exactly as in the source. The truncation is also minimal for non-integer p : the first surviving term after substitution $t = \alpha|u|^q$ has power $|u|^{qk_p}$, and

$$qk_p \geq p, \quad q(k_p - 1) < p.$$

Thus $W^{1,p} \subset L^p$ controls the surviving small values, while deleting one fewer term would in general require an unavailable L^r estimate with $r < p$.

An isoperimetric theorem for every real exponent

Let $\mathcal{I}$ be the isoperimetric profile of an infinite-volume metric measure space.

Theorem 1. *Let $p > 1$. Suppose (X, d, μ) is locally compact and geodesic and there are $c, h, s_0 > 0$ such that*

$$\mathcal{I}(s) \geq c s^{1-1/p} \quad (0 < s \leq s_0), \quad \mathcal{I}(s) \geq h s \quad (s \geq s_0). \quad (1)$$

Then there is a constant $C = C(p, c, h, s_0)$ such that every $u \in W^{1,p}(X)$ satisfying

$$\int_X \text{lip}(u)^p d\mu \leq 1$$

also satisfies

$$\int_X \mathcal{E}_p(\alpha|u|^q) d\mu \leq C \quad \text{for every } 0 < \alpha \leq c^q. \quad (2)$$

Proof. Let $v = u^* : (0, \infty) \rightarrow [0, \infty)$ be the decreasing rearrangement of $|u|$ in the measure variable. The coarea–Hölder argument used in the source gives, for every real $p > 1$,

$$\int_0^\infty |v'(s)|^p \mathcal{I}(s)^p ds \leq \int_X \text{lip}(u)^p d\mu \leq 1. \quad (3)$$

Approximation gives the same statement for general $W^{1,p}$ functions. Equimeasurability reduces (2) to an integral involving v .

The two bounds in (1) together give the global linear estimate $\mathcal{I}(s) \geq \bar{h}s$, where

$$\bar{h} = \min\{h, cs_0^{-1/p}\}.$$

Thus the standard Cheeger–Poincaré estimate gives

$$\|v\|_{L^p(0,\infty)} = \|u\|_{L^p(X)} \leq \frac{p}{\bar{h}} \|\text{lip}(u)\|_{L^p(X)} \leq \frac{p}{\bar{h}}.$$

In particular, v is uniformly bounded on $[s_0, \infty)$. Since $\mathcal{E}_p(t) = O(t^{k_p})$ at zero and $qk_p \geq p$, it follows that

$$\int_{s_0}^\infty \mathcal{E}_p(c^q v(s)^q) ds \leq C_1(p, c, h, s_0). \quad (4)$$

It remains to treat $0 < s < s_0$ at the endpoint. Set $B = v(s_0)$ and

$$\begin{aligned} e_0 &= c^p \int_0^{s_0} |v'(s)|^p s^{p-1} ds, \\ e_1 &= h^p \int_{s_0}^\infty |v'(s)|^p s^p ds. \end{aligned}$$

By (1)–(3), $e_0 + e_1 \leq 1$. Since $v(s) \rightarrow 0$ as $s \rightarrow \infty$, weighted Hölder gives

$$(cB)^p \leq D e_1, \quad D = \frac{c^p}{h^p} \frac{(p-1)^{p-1}}{s_0}. \quad (5)$$

Indeed, integrate $-v'$ from s_0 to infinity and use $\int_{s_0}^{\infty} s^{-q} ds = (p-1)s_0^{-1/(p-1)}$.

Use the logarithmic coordinate $t = \log(s_0/s)$ and define

$$w(t) = c(v(s_0 e^{-t}) - B), \quad a = cB.$$

Then $w(0) = 0$, w is nondecreasing, and

$$\int_0^{\infty} |w'(t)|^p dt = e_0, \quad a^p \leq D(1 - e_0).$$

We use the following shifted form of Moser's one-dimensional lemma.

Lemma 1. *For $p > 1$, $q = p/(p-1)$ and $D > 0$, there is $M(p, D) < \infty$ such that*

$$\int_0^{\infty} e^{(w(t)+a)^q - t} dt \leq M(p, D)$$

whenever $w(0) = 0$, $w' \geq 0$, $\int |w'|^p = \rho \leq 1$, and $a^p \leq D(1 - \rho)$.

To check the lemma, if $0 < \rho < 1$ put $z = \rho^{-1/p}w$. The elementary maximization over $x \geq 0$ gives

$$(\rho^{1/p}x + a)^q - x^q \leq \left(\frac{a^p}{1 - \rho} \right)^{1/(p-1)} \leq D^{1/(p-1)}. \quad (6)$$

Moser's one-dimensional lemma

$$\sup_{\substack{z(0)=0, z' \geq 0 \\ \int_0^{\infty} |z'|^p \leq 1}} \int_0^{\infty} e^{z(t)^q - t} dt < \infty$$

then proves the claim. The cases $\rho = 0$ and $\rho = 1$ follow directly (for $\rho = 1$, the hypothesis forces $a = 0$). The classical proof uses only Hölder's inequality and is valid for every real $p > 1$ [2].

Finally, $ds = s_0 e^{-t} dt$ and Lemma 1 imply

$$\int_0^{s_0} \mathcal{E}_p(c^q v(s)^q) ds \leq s_0 \int_0^{\infty} e^{(w(t)+a)^q - t} dt \leq s_0 M(p, D).$$

Together with (4), this proves the endpoint $\alpha = c^q$; smaller α follow by monotonicity. $\square$

The non-integer curvature-dimension corollary

Theorem 2. *Let $N > 1$ be real, and let (X, d, μ) be an infinite-volume, locally compact, geodesic, essentially nonbranching $\text{CD}(K, N)$ space. Assume*

$$\inf_{x \in X} \mu(B(x, 1)) > 0 \quad \text{and} \quad h(X) > 0.$$

Then there are $\alpha_N, C_N > 0$ such that, for every $u \in W^{1,N}(X)$,

$$\int_X \text{lip}(u)^N d\mu \leq 1 \quad \implies \quad \int_X \left[e^{\alpha_N |u|^{N/(N-1)}} - \sum_{j=0}^{[N-1]-1} \frac{\alpha_N^j |u|^{jN/(N-1)}}{j!} \right] d\mu \leq C_N.$$

Proof. The small-volume isoperimetric theorem quoted and used in the source is valid for real curvature parameter N : the unit-ball lower bound supplies $c, s_0 > 0$ with

$$\mathcal{I}(s) \geq c s^{1-1/N} \quad (0 < s \leq s_0).$$

The positive Cheeger constant supplies $\mathcal{I}(s) \geq h(X)s$. Apply Theorem 1 with $p = N$ and $\alpha_N = c^{N/(N-1)}$. $\square$

This answers the source's question affirmatively under precisely the two geometric nondegeneracy hypotheses used for its integer-dimensional positive theorem. No fractional differential operator is introduced: fractional dimension refers to the real curvature-dimension parameter and the corresponding real Sobolev exponent.

Deep upgrade checks and scope

1. The first proof attempt gave only $\alpha < c^q$ by the crude estimate $v(s) \lesssim (\log(1/s))^{1/q}$. The tail-energy estimate (5) and the exact algebraic bound (6) upgrade this to the endpoint $\alpha = c^q$.
2. The ceiling in $\mathcal{E}_p$ is forced, not cosmetic: it is the unique minimal truncation that makes the first remainder power at least p .
3. The argument needs a small-ball lower bound and a positive Cheeger constant. The source itself shows why analogous hypotheses are necessary in its integer formulation; the result does not claim that every $\text{CD}(K, N)$ space satisfies a global inequality.
4. The constant c is any uniform coefficient in the small-volume isoperimetric inequality. Identifying the sharp largest coefficient from a mere liminf profile may require additional regularity and is not asserted here.

Bounded novelty check

On 11 August 2026, non-integer $\text{CD}(K, N)$ Moser–Trudinger inequalities, fractional-dimensional weighted Sobolev inequalities, real-dimensional radial measures, and isoperimetric-profile proofs were checked in the run indexes and by bounded web/arXiv search. Weighted one-dimensional fractional-dimension inequalities are known [3], but no direct answer to the source’s $\text{CD}(K, N)$ question or the profile theorem above was located. This is evidence for novelty, not a definitive guarantee.

References

- [1] S. Bronstein, *Dominating manifolds by radial spaces*, arXiv:2506.15621.
- [2] J. Moser, *A sharp form of an inequality by N. Trudinger*, Indiana Univ. Math. J. **20** (1971), 1077–1092.
- [3] J. F. de Oliveira and J. M. do Ó, *Equivalence of critical and subcritical sharp Trudinger–Moser inequalities in fractional dimensions and extremal functions*, Rev. Mat. Iberoam. **39** (2023), 1073–1096.